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A Note on Local Tax Rates, Public Benefit Levels, and Property Values

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I. Introduction

Within the past decade several efforts have been made to investigate econometrically the extent to which fiscal differentials between communities are capitalized into property values (see, e.g., Orr 1968; Oates 1969, 1973; Church 1974; Meadows 1976). Economic theory suggests that under certain conditions, if a community has higher tax rates than average, its property values will be lower, other things being equal (see Polinsky and Shavell 1976). Similarly, local public services "better" than average should result in higher property values.[1]

In several important studies of tax capitalization (Orr 1968; Oates 1969), expenditures on various local public benefits have been used to proxy the quantity and quality of local public services. It is widely recognized that such measures are deficient because of the untenable assumption that output can be measured by expenditures on inputs. The purpose of this essay is to present some empirical results on tax capitalization using data on *output* measures to correct for the influence of local public benefits on property values. We find that when such information is included in the analysis, the statistical results are more consistent with theoretical predictions than when expenditure levels are used. In particular, our results suggest capitalization rates close to 90 percent.

In Section II the econometric literature on property tax capitalization is discussed briefly, with special attention focused upon the treatment of public benefits. In Section III we estimate the parameters of a tax-

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[1] Polinsky and Shavell (1976) show that the use of cross-section regression results to predict property value adjustments in response to "amenity" changes is valid when the communities considered are "small" and there is mobility within and among different areas. The sample of communities studied in this paper seems to approximate these conditions fairly well.

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capitalization model, contrasting the results when input rather than output measures of local benefits are used. The implied capitalization ratio is then calculated. In a concluding section we discuss some limitations of the present analysis and suggestions for future research.

II. Public Benefits Levels in Capitalization Studies

In several studies, investigators have attempted to examine the question of tax capitalization without considering the public benefits financed by the taxes (e.g., Smith 1970).[2] However, it would seem desirable to examine the capitalization issue in a framework that explicitly accounts for all the nontax determinants of housing values, including public service levels. The best-known example of such an approach is Oates's (1969) important cross-section study of capitalization in 53 northeastern New Jersey communities for the year 1960. Oates regresses the median value of housing on the natural logarithm of the effective property tax rate, the logarithm of per pupil expenditure,[3] and a set of variables selected to correct for the quality of housing. In Section III below we will reproduce Oates's estimates; we confine ourselves now to noting his important qualitative results. These are that the tax variable has a significant negative sign, and the expenditure-per-pupil variable has a significant positive sign.[4] Oates shows that his regression coefficients imply a tax-capitalization ratio of about two-thirds.[5]

As Oates points out, a number of problems are associated with this approach, particularly in specifying the nontax variables that determine housing value (1969, p. 176). However, here we will focus mainly upon the public benefits measure, expenditure per pupil. Although property values probably are influenced by the quality of schooling, it is not at all clear that the latter is measured adequately by expenditure per pupil. There are several reasons why this might be the case: (*a*) educational factor prices might differ among communities, (*b*) educational production

[2] The purpose here is not to provide a comprehensive review of the empirical literature on tax capitalization. (Church [1974] provides a discussion of several of the studies.) Rather, our goal is to give a general view of how local public benefits levels have been treated previously. For a discussion of some of the theoretical issues, see Mieszkowski (1972).

[3] Orr (1968) also includes a measure of community public services in his study. He uses annual educational expenditure per pupil in public schools expressed as a deviation from the state average. For a critique of Orr's method, see Coen and Powell (1972). Church (1974) and Wales and Wiens (1974) try to control for the influence of public service differentials by limiting their sample to microeconomic data from only one district.

[4] These results obtain under both ordinary least squares and two-stage least squares.

[5] This calculation is done for a $20,000 house with a life of 40 years. The discount rate is assumed to be 5 percent.

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functions might differ among communities, and (*c*) endowments of non-market inputs might differ among communities.[6] The main point is that the value of inputs is not likely to be a good measure of output in this context.

What we seek, then, is a measure of perceived quality of local public benefits. Recently, student achievement test scores for the communities used in the Oates study have become available.[7] We propose to estimate the housing value equations using these test score data rather than expenditure per pupil. Clearly, the test scores are not a perfect measure. For example, the expected "value added" by the school system may be a more important factor than its overall quality. More importantly, examination results certainly cannot be guaranteed to summarize adequately the whole vector of characteristics that determine perceived school quality. Nevertheless, the use of test scores probably represents a significant improvement over input variables as a measure of public benefits.

III. Results

We have argued that the outputs of public services and amenities should be held constant in studies of tax capitalization, not expenditures on inputs. In this section we use data from 1970 to estimate Oates's model using output instead of input measures of public benefits. Since Oates's study employed 1960 data, we first reestimate it with our more recent data for the same sample of New Jersey communities in order to facilitate direct comparisons.

Updating the Oates Model

The basic model to be estimated is:

$$V = \alpha_0 + \alpha_1 \ln t + \alpha_2 \text{PUB} + \alpha_3 \ln D + \alpha_4 R + \alpha_5 N + \alpha_6 Y + \alpha_7 P + u, \tag{1}$$

[6] More formally, assume an educational production function $Q = f(M, E)$, where Q is the quality adjusted amount of educational services, M is a vector of market inputs, and E is a vector of "environmental" factors that influence school quality, e.g., social background of the students. It is possible for a community with sufficiently low values of the E variables to spend more on market inputs and yet have a lower quality of education. See Oates (1976).

[7] We would like to thank the late Professor Frederick Harbison for making these data available to us. They were not available for three of the communities in the original Oates sample.

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where[8]

V = median value of owner-occupied housing in thousands of dollars;
t = average effective tax rate over the preceding 5 years;[9]
PUB = some measure of public benefit levels;
D = distance to midtown Manhattan;
R = median number of rooms per home;
N = percentage of housing stock built in the last 10 years;
Y = median income of individuals in thousands of dollars;
P = percentage of the population that has income under $4,000 (measured in 1970 dollars);
u = random error term.

The rationalization for this particular functional form is discussed at length by Oates; we merely note that none of the results described below seems to be very sensitive to changes in functional form.

In the Oates model, PUB is equal to educational expenditures per pupil (EXP).[10] As Oates notes, there is likely to be correlation between the error term and the expenditure and tax variables. In order to deal with the simultaneity problem, Oates estimates (1) by two-stage least squares (2SLS);[11] we follow the same procedure in our analysis of the 1970 data.

The estimated parameters of (1) for both 1960 and 1970 are shown in table 1. (All dollar amounts are in terms of 1970 dollars.) Ignoring for the moment the expenditure variable, we see that regardless of the year and regardless of estimation technique, the coefficients have signs as expected. Property values increase with proximity to the central city, size and newness of houses, and with the incomes of their inhabitants. The

[8] Data on housing values and number of rooms are from the U.S. Bureau of the Census (1970*a*). Taxes are calculated from data published by the New Jersey Taxpayer's Association (1970). Data on population, percentage poor, educational attainment, and income are from the U.S. Bureau of the Census (1970*b*). Commercial and industrial property figures are from the State of New Jersey, Department of Community Affairs (1971).

[9] The effective tax rate is the assessment ratio times the nominal tax rate. In the equations presented below, the effective tax rate always performs better than the nominal tax rate, as economic theory would suggest.

[10] Oates used a weighted average of enrollments to calculate the expenditure-per-student variable in order to account for increased costs of special students and students in higher grades. The weighted expenditure variable was calculated by dividing current costs by the weighted enrollment figure. He weighted kindergarten students as 0.5, students in grades 1–8 at 1.0, students in grades 9–12 as 1.25, and special students as 2.0.

[11] The instruments used are median number of years of school completed by males of age 25 or more, population density, percentage of dwellings owner occupied, percentage change in population over the past decade, percentage enrolled in public elementary and secondary schools, a dummy variable for Hudson County, the value of commercial and industrial property per resident.

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TABLE 1

DETERMINANTS OF MEDIAN HOUSING VALUES FOR 53 NEW JERSEY COMMUNITIES, PUBLIC BENEFITS MEASURED BY EXPENDITURE PER PUPIL

Estimation technique	OLSQ	2SLS	OLSQ	2SLS
Year	1960	1960	1970	1970
Constant	−28	−42	−17	7.0
	(12)	(18)	(11)	(26)
ln t	−4.6	−4.4	−7.5	−7.6
	(1.2)	(1.5)	(1.7)	(3.0)
ln EXP	4.2	6.5	2.1	−1.6
	(2.0)	(3.1)	(1.6)	(4.1)
ln D	−1.7	−1.6	−1.7	−2.1
	(0.39)	(0.42)	(0.57)	(0.72)
R	2.4	2.2	1.9	2.2
	(0.56)	(0.60)	(0.96)	(1.4)
N	0.07	0.07	0.05	0.04
	(0.02)	(0.02)	(0.03)	(0.03)
Y	1.5	1.4	2.2	2.2
	(0.18)	(0.19)	(0.22)	(0.24)
P	0.4	0.4	0.5	0.5
	(0.11)	(0.12)	(0.17)	(0.20)
R^2	.93	.93	.92	.91

NOTE.—Numbers in parentheses are standard errors. Variables are defined in the text.

usual analysis-of-variance tests applied to the ordinary least-squares results for 1960 and 1970 suggest that the hypothesis that this set of regression coefficients has remained unchanged cannot be rejected at standard significance levels.

However, the expenditure-per-pupil variable for 1970 does not perform very well. Not only is it statistically insignificant, but in the two-stage least-squares regression it has a negative sign. This suggests that the per pupil expenditure variable is inadequate as a measure of public benefits. For the reasons discussed in Section II above, this comes as no great surprise.

It could be argued that the weighting scheme used to compute expenditure per pupil is inappropriate, and that the expenditure variable might have had a positive sign had a better measure been used. In order to test for this possibility, (1) was reestimated using a number of alternative measures of EXP: an unweighted sum of per pupil expenditure, a measure of funds spent on student body activities, and a measure which excludes certain expenditures that might not increase the perception of school quality (e.g., payments made to other school districts). All of these measures had coefficients with negative statistically insignificant signs in the 1970 two-stage least-squares regression.[12] Thus, per pupil expenditure, no matter how defined, does not capture adequately the impact of the quality of education on local property values.

[12] These results are available upon request to the authors.

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Achievement Test Scores as Measures of Local Public Benefits

Before presenting the results when achievement test scores are substituted for PUB in (1), we should again emphasize that whether or not they reflect actual school quality is not the issue. The question is whether or not they represent perceived quality of education better than do expenditure data.

The test score data consist of the average grades for fourth-grade pupils on reading and mathematics examinations in each community. Since theory does not indicate how the test scores should enter the equation, we estimated (1) several times, each with a different definition of the PUB variable: RD (reading score), MA (math score), TS (average of reading and math score), and LTS (logarithm of the average test score). The estimation technique was two-stage least squares, with instruments the same as those listed above. The results are shown in table 2. In each column the third entry gives the coefficient of the public benefit measure which heads the column. Thus, for example, .50 is the coefficient of the reading score variable (RD).

Regardless of how the test scores enter the equation, they have positive coefficients that differ significantly from zero at the 95 percent level. The coefficients on all the other variables are of the expected signs, and the relative sizes of the standard errors are smaller than those found in table 1. Note in particular that the estimated coefficient of the tax variable in each equation is negative and exceeds its standard error by at least a factor of three. Substitution of the test scores for per pupil expenditure thus seems to improve the estimates. It is difficult to judge the "reasonableness" of the coefficients on the test score variables. The range of the reading score variable is 47.08–64.04. The top decile is marked by a score of 63.27, the bottom decile by 54.59. According to our estimates, then, being in the first rather than the last decile would increase median property values by about $4,300, other things being equal.

Capitalization

We now turn to the problem of calculating the capitalization ratio implied by our regression coefficients. Consider a $30,000 house with a 40-year use life, and assume a discount rate of 6 percent. Full capitalization of a change in the effective tax rate from 3 to 4 percent would induce a downward change in the value of the house of about $2,820. The coefficient of the tax variable on the first equation in table 2[13] implies that

[13] Of course, these figures would vary under alternative sets of initial conditions. In a set of regressions not reported here, crime rate variables were included as well as test scores. In these equations, the tax rate coefficients were generally even higher than those reported in table 2. However, the coefficients on the crime rate variables themselves were statistically insignificant.

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TABLE 2

DETERMINANTS OF MEDIAN HOUSING VALUES FOR 50 NEW JERSEY COMMUNITIES, PUBLIC BENEFITS MEASURED BY ACHIEVEMENT TEST SCORES

Independent Variables	RD	MH	TS	LTS
Constant	−32	−20	−28	−45
	(9.0)	(7.1)	(8.2)	(13)
ln t	−8.6	−8.0	−8.5	−8.2
	(2.1)	(2.2)	(2.1)	(2.1)
RD, MH, TS, LTS	0.50	0.27	0.42	23.1
	(0.16)	(0.11)	(0.14)	(7.9)
ln D	−2.4	−2.0	−2.1	−2.1
	(0.54)	(0.54)	(0.53)	(0.53)
R	2.4	2.0	2.3	2.2
	(1.0)	(1.0)	(1.0)	(0.98)
N	0.09	0.07	0.08	0.08
	(0.03)	(0.03)	(0.03)	(0.03)
Y	1.9	2.1	2.0	2.0
	(0.24)	(0.24)	(0.24)	(0.23)
P	0.8	0.7	0.8	0.8
	(0.18)	(0.19)	(0.19)	(0.19)
R^2	.94	.93	.94	.93

NOTE.—Estimation technique is two-stage least squares. Numbers in parentheses are standard errors, and variables are defined in the text. Dependent variable is median value of housing in thousands of dollars.

such a change would reduce the value by about $2,500. Thus, about 88 percent of the tax differential is capitalized. A similar calculation using the regression which controlled for public benefit levels using per pupil expenditure (from the fourth column of table 1) yields a capitalization ratio of only about 75 percent. Thus the capitalization ratio increases considerably when an output rather than an expenditure measure of local public benefits is used.[14]

IV. Concluding Remarks

In this note attention has been focused upon the measurement of local public benefits in econometric studies of tax capitalization. We have stressed that it is inappropriate to use variables such as per pupil expenditure, because inputs should not be used to measure outputs. Indeed, using 1970 data we showed that per pupil expenditure performs unsatisfactorily in a property value equation. On the other hand, the use of school achievement scores, although an imperfect measure of perceived educational quality, improves the equation considerably. Our results imply that the bulk of property tax differentials are capitalized into local property values.

These results must be viewed with some caution. Our estimates may be sample specific. Other areas might differ with respect to information on

[14] Unfortunately, it is not possible to predict the extent to which a tax-induced change in housing values would be offset by improved quality of schools. This is because we have no usable information on the educational production function.

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effective tax rates and public benefit levels. Differences in zoning regulations would also be likely to make a difference. Finally, although we have experimented with a number of alternative variables and functional forms, there may nevertheless be specification errors present.

However, the most important limitation of this study is its partial equilibrium framework. Recent advances in the theory of property tax incidence (Mieszkowski 1972; Aaron 1975) emphasize that the out-migration of capital from the housing sector drives down the rate of return to capital in the rest of the economy. Therefore, calculation of a capitalization ratio assuming a fixed discount rate may be misleading. It remains an open question whether or not the general equilibrium model is amenable to econometric testing.

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